

Lecture 2 - Vector norms, orthogonality, eigenvalues/eigenvectors

Norms $\rightsquigarrow$ "length" of vectors

Example:

$$u = \begin{bmatrix} 2 \\ -2 \\ 2 \\ 2 \end{bmatrix}$$

$$v = \begin{bmatrix} 0 \\ 4 \\ 0 \\ 0 \end{bmatrix}$$

$$\|u\|_2 = \sqrt{4+4+4+4} = 4$$

$$\|u\|_1 = 8$$

$$\|u\|_\infty = 2$$

$$\|v\|_2 = \|v\|_1 = \|v\|_\infty = 4$$

Def: A function $\|\cdot\| : \{\text{vectors}\} \rightarrow \mathbb{R}$ is a vector norm if

1) $\|v\| \geq 0$ for all v and $\|v\| = 0 \Leftrightarrow v = 0$

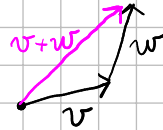
2) $\|\alpha \cdot v\| = |\alpha| \cdot \|v\| \quad \forall \alpha \in \mathbb{R} \quad \forall v \text{ vector}$

3) $\|v+w\| \leq \|v\| + \|w\| \quad \forall v, w$ TRIANGLE INEQUALITY

Examples:

• Euclidean norm: $v \in \mathbb{R}^n \quad v = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$

$$\|v\|_2 = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2} = \sqrt{v^T v} = \sqrt{\langle v, v \rangle}$$



• $\|v\|_1 = |v_1| + |v_2| + \dots + |v_n|$

• $\|v\|_{\max} = \max_{i=1, \dots, n} |v_i|$

Fact: for any vector $v \neq 0$, you can write $v = \alpha \cdot w$

$\alpha = \frac{v}{\|v\|}$ (unit vector)

ORTHOGONALITY

Def: A matrix $U \in \mathbb{R}^{n \times n}$ is orthogonal if $U U^T = U^T U = I$

Equivalently: $U = \begin{bmatrix} | & | & | \\ u_1 & u_2 & \dots & u_n \\ | & | & | \end{bmatrix}$ $\{u_1, u_2, \dots, u_n\}$ are an orthonormal basis of $\mathbb{R}^n$

Def: Two vectors u, v are

orthogonal to each other if $u^T v = 0$.

$$\langle u_i, u_j \rangle = u_i^T u_j = \begin{cases} 1 & \text{if } i=j \\ 0 & \text{if } i \neq j \end{cases}$$

Examples:

I_n

$$U = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \quad \theta \in [0, 2\pi]$$

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \mapsto U \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \cos \theta \\ -\sin \theta \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} \mapsto U \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \sin \theta \\ \cos \theta \end{bmatrix}$$

$$U^T U = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Properties: For any matrices A, B "that make sense"

$$1. (AB)^T = B^T A^T \rightsquigarrow \boxed{}$$

$$2. (AB)^{-1} = B^{-1} A^{-1}$$

$$\hookrightarrow \underbrace{B^{-1} A^{-1}}_I (AB) = B^{-1} B = I$$

Lemma: If U_1, U_2 are orthogonal matrices of size $n \times n$

then $\underbrace{U_1 U_2}$ is also an orthogonal matrix.

$$(U_1 U_2)^T (U_1 U_2) = U_2^T \underbrace{U_1^T U_1}_I U_2 = U_2^T U_2 = I$$

Property: If $U \in \mathbb{R}^{n \times n}$ is orthogonal and $x \in \mathbb{R}^n$

then $\|Ux\| = \|x\|$

Proof: $\|Ux\| = \sqrt{(Ux)^T (Ux)}$ Euclidean norm.

$$= \sqrt{x^T \underbrace{U^T U}_I x} = \sqrt{x^T x} = \|x\| \quad \square$$

Sometimes, it's useful to consider $U \in \mathbb{R}^{n \times k}$ $\begin{bmatrix} | & | & | \\ u_1 & u_2 & u_3 \\ | & | & | \end{bmatrix}$

with orthonormal columns,

that is, its columns satisfy $u_i^T u_j = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}$

Q: $U^T U = \boxed{} \boxed{} = I_k \quad (U^T U)_{ij} = \langle u_i, u_j \rangle$

$$U U^T = \boxed{} \boxed{}$$

$\hookrightarrow \text{rank} = k \Rightarrow \text{cannot be identity}$

$$U U^T = u_1 u_1^T + u_2 u_2^T + \dots + u_k u_k^T$$

Note: There is no matrix $U = \boxed{} \boxed{} \boxed{} \boxed{}$ with orthonormal columns

Q: If $U \in \mathbb{R}^{n \times k}$ has orthonormal columns, $x \in \mathbb{R}^k$, $y \in \mathbb{R}^n$

(a) $\|Ux\| = \|x\|$? $\checkmark$

~~$\|U^T y\| = \|y\|$? $\times$~~

$$\boxed{} \quad \|Ux\| = \sqrt{x^T \underbrace{U^T U}_I x} = \sqrt{x^T x} = \|x\|$$

$$\boxed{} \quad \exists v \neq 0 \text{ s.t. } U^T v = 0$$

EIGENVALUES AND EIGENVECTORS

Def: $A \in \mathbb{R}^{n \times n}$, λ is eigenvalue of A , $v \neq 0$ a corresponding eigenvector if $Av = \lambda v$.

"Most" matrices are diagonalizable: $\exists V \in \mathbb{R}^{n \times n}$ invertible, $D = \begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \\ & & & \lambda_n \end{bmatrix}$

such that $A = V \cdot D \cdot V^{-1}$

$$= \begin{bmatrix} | & | & | \\ v_1 & v_2 & \dots & v_n \\ | & | & | \end{bmatrix} \begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \\ & & & \lambda_n \end{bmatrix} \begin{bmatrix} -w_1^T- \\ -w_2^T- \\ \vdots \\ -w_n^T- \end{bmatrix} \quad \begin{array}{l} (\lambda_i, v_i) \text{ eigenpairs of } A \\ \{v_1, v_2, \dots, v_n\} \text{ basis of } \mathbb{R}^n \\ \text{made of eigenvectors of } A \end{array}$$

$$= \underbrace{\lambda_1 v_1 w_1^T + \lambda_2 v_2 w_2^T + \dots + \lambda_n v_n w_n^T}_{\text{rank-1 matrix}}$$

Examples: $A = \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix}$ $A \cdot e_1 = 2e_1$ $A \cdot e_2 = -e_2$
 $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ $A \cdot x = x$ all vectors $\neq 0$ are eigenvectors corresponding to the eigenvalue 1

Eigenvalues are the zeros of the characteristic polynomial:

$$\det(A - \lambda I) = 0$$

Matlab: `qr` $\leadsto$ gives you

all eigenpairs in $O(n^3)$

(not using characteristic poly)

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \leadsto \det \begin{bmatrix} -\lambda & -1 \\ 1 & -\lambda \end{bmatrix} = \lambda^2 + 1$$

$\hookrightarrow$ 2 complex eigenvalues

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \leadsto (1-\lambda)^2 \leadsto \text{cannot build a basis of eigenvectors}$$

Rmk: Eigenvalues of A are unique (up to reordering)

Eigenvectors not unique: v eigenvector $\Rightarrow \alpha v$ eigenvector
 $Av = \lambda v$ $A(\alpha v) = \alpha(Av) = \alpha \lambda v$

v, w eigenvectors corresp. to λ

then $v+w$ eigenvector corr to λ

$$A(v+w) = Av + Aw = \lambda v + \lambda w = \lambda(v+w)$$

Q: $A = VDV^{-1}$ $\rightarrow$ eigendecomposition $\rightarrow$ eigenvalues and eigenvectors of A^k ?

$$A^k = A \cdot A \cdot \dots \cdot A = V \cancel{DV^{-1}} \cancel{VDV^{-1}} \dots \cancel{VDV^{-1}} = \underline{VD^kV^{-1}}$$

$$D^k = \begin{bmatrix} \lambda_1^k & & \\ & \lambda_2^k & \\ & & \ddots \\ & & & \lambda_n^k \end{bmatrix}$$

The eigenvalues are $\lambda_1^k, \dots, \lambda_n^k$

The eigenvectors are the same as A.

$$p(x) = c_0 + c_1x + c_2x^2 + \dots + c_kx^k$$

$$p(A) = c_0I + c_1A + c_2A^2 + \dots + c_kA^k$$

$$= V \cdot (c_0I + c_1D + c_2D^2 + \dots + c_kD^k) V^{-1} = V \cdot p(D) \cdot V^{-1}$$

$$p(D) = \begin{bmatrix} p(\lambda_1) & & \\ & p(\lambda_2) & \\ & & \ddots \\ & & & p(\lambda_n) \end{bmatrix}$$

Def: $A \in \mathbb{R}^{n \times n}$ is SYMMETRIC if $A^T = A$



SPECTRAL THEOREM: If $A \in \mathbb{R}^{n \times n}$ symmetric $\exists U \in \mathbb{R}^{n \times n}$ orthogonal matrix and $D = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix} \in \mathbb{R}^{n \times n}$ such that

$$A = UDU^T = \boxed{U} \boxed{\begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}} \boxed{U^T}$$

• eigenvalues are real

• we have n eigenvectors that form an orthonormal basis of $\mathbb{R}^n$.

Theorem: $A \in \mathbb{R}^{n \times n}$ symm $\Rightarrow \lambda_{\min} \|x\|^2 \leq \underbrace{x^T A x}_{\text{quadratic form}} \leq \lambda_{\max} \|x\|^2$ for any x

Proof: • Simple case first:

assume that $A = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}$ $\lambda_1 = \lambda_{\max} \geq \lambda_2 \geq \dots \geq \lambda_n = \lambda_{\min}$

$$\begin{aligned} x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} &\rightarrow x^T A x = \lambda_1 x_1^2 + \lambda_2 x_2^2 + \dots + \lambda_n x_n^2 \quad \otimes \\ &\leq \lambda_1 x_1^2 + \lambda_1 x_2^2 + \dots + \lambda_1 x_n^2 \\ &= \lambda_1 (x_1^2 + x_2^2 + \dots + x_n^2) = \lambda_1 \|x\|^2 \quad \checkmark \end{aligned}$$

$$\otimes \geq \lambda_n x_1^2 + \lambda_n x_2^2 + \dots + \lambda_n x_n^2 = \lambda_n \|x\|^2 \quad \checkmark$$

• General case: by spectral theorem, $A = UDU^T$

$$x^T A x = \underbrace{x^T U}_{\text{orthonormal}} \underbrace{DU^T}_{\text{orthonormal}} x$$

By the "easy case" applied to the vector $U^T x$,

$$\lambda_{\min} \cdot \underbrace{\|U^T x\|^2}_{\|x\|^2} \leq x^T U D U^T x \leq \lambda_{\max} \cdot \underbrace{\|U^T x\|^2}_{\|x\|^2}$$

□

Remark: if $v \neq 0$ s.t. $Av = \lambda_{\max} \cdot v$

$$v^T \underbrace{A} v = v^T \cdot (\lambda_{\max} \cdot v) = \lambda_{\max} \cdot v^T v = \lambda_{\max} \|v\|^2$$

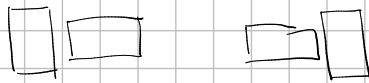
same thing for $\lambda_{\min}$...

Def: $A \in \mathbb{R}^{n \times n}$ is SYMMETRIC POSITIVE DEFINITE (SPD) if all eigenvalues are strictly positive. A is SYMMETRIC POSITIVE SEMIDEFINITE (SPSD) if all the eigenvalues are ≥ 0 .

Equivalently: A SPD $\Leftrightarrow x^T A x > 0$ for all $x \neq 0$ ←

A SPSPD $\Leftrightarrow \underline{x^T A x \geq 0}$ for all x

Note: if $A \in \mathbb{R}^{n \times m}$ (possibly rectangular),
the matrices AA^T and $\underbrace{A^T A}$ are SPSPD.



Let's show that AA^T is SPSPD

• symmetry: $(AA^T)^T = (A^T)^T A^T = AA^T$ ✓

• $x \in \mathbb{R}^n$ $x^T AA^T x \stackrel{?}{\geq} 0$

$$\hookrightarrow = (A^T x)^T \cdot (A^T x) = \|A^T x\|^2 \geq 0$$

$\Rightarrow AA^T$ is SPSPD.

In the complex case:

$$A \in \mathbb{C}^{n \times n}$$

• replace T with "transpose + entrywise conjugate" A^H, A^*

• $U \in \mathbb{C}^{n \times n}$ unitary if $U^* U = U U^* = I$ ("like orthogonal")

• A is Hermitian if $A^* = A$ ("like symmetric")